

MTE544 – Autonomous Mobile Robotics

University of Waterloo

Department of Mechanical and Mechatronics Engineering

Lab 1: Sensor data processing for mobile robots

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Station number: _____

Robot number: 7

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1 Odometry results and discussion

Figure 1 retains all logged poses. For each file, $t = (s - s_0)10^{-9}$ s converts nanosecond stamps to elapsed time. Positions remain in the odometry frame; yaw is unwrapped to remove 2π representation jumps.

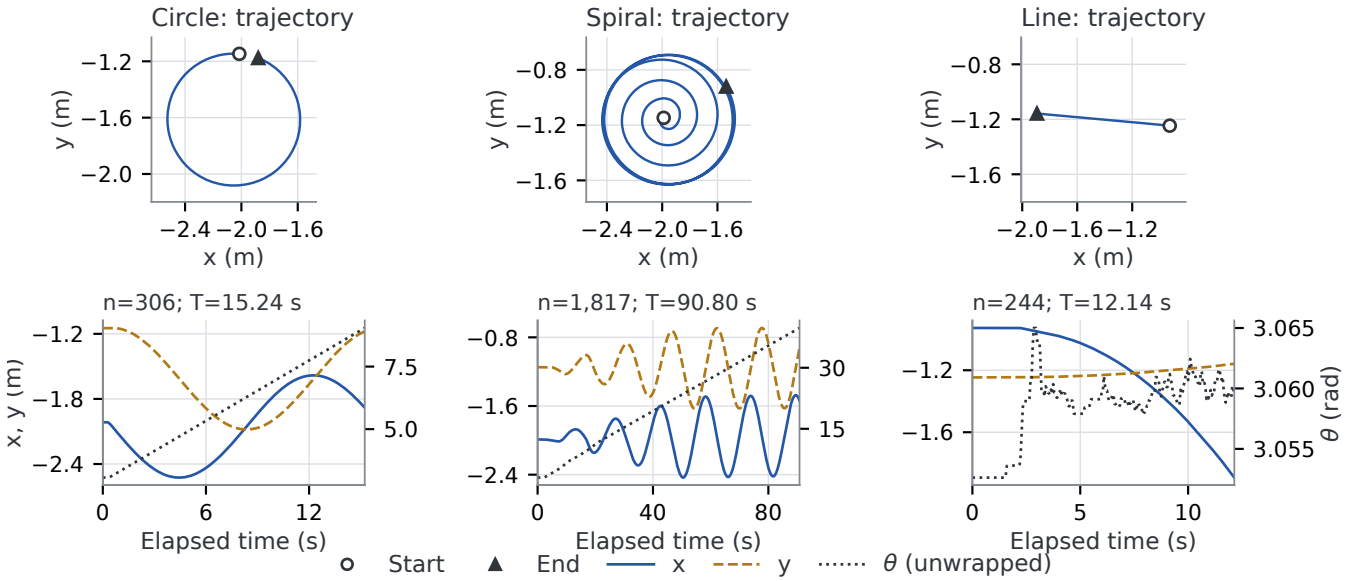


Figure 1: Odometry: equal-aspect x–y paths (top) and position/heading histories (bottom). Left axes show metres; right axes show unwrapped radians. Each run uses its own axis ranges. Titles give sample count n and duration T.

Circle: The resulting radius is 0.470 m; θ increased by 5.98 radians during the 15.24 s recording. Since this is less than 2π radians, a full circle was not completed.

Spiral: The outer-loop radius is approximately 0.470 m; θ increased by 36.79 radians during the 90.80 s recording. The loops initially widen, then overlap near the outer boundary as the radius becomes approximately constant.

Line: The path length is 0.970 m over 12.14 s, with heading variation of only 0.0124 rad (0.71°). The increasing magnitude of the position slope indicates acceleration.

2 IMU results and discussion

Figure 2 shows logged IMU-frame accelerations (specific force) and z-axis angular rate.

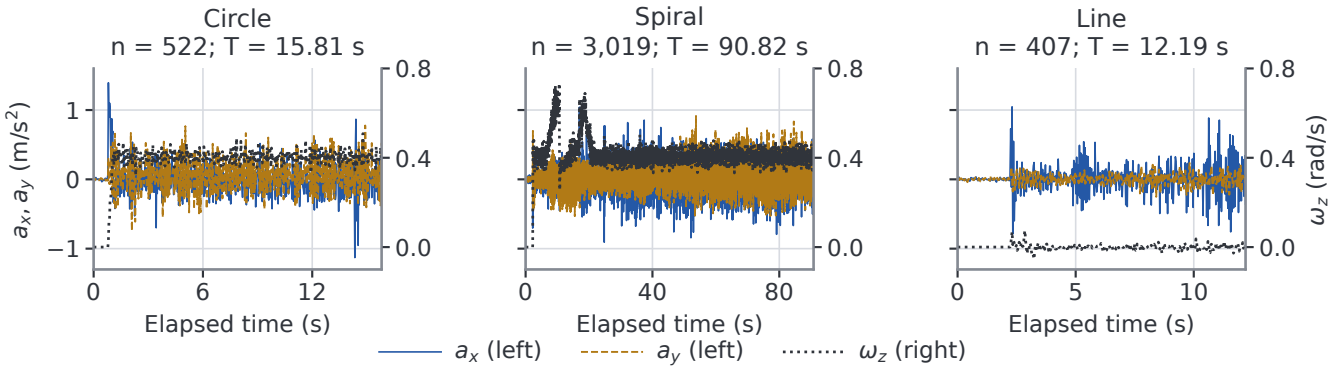


Figure 2: Raw IMU samples on shared scales; acceleration uses left axes and angular rate right. Fluctuations may include motion and measurement effects.

Circle: Over 15.81 s, ω_z settles to 0.402 rad/s (mean over 2–15 s). Full-run mean $a_y = 0.0594$ m/s^2 is consistent with inward acceleration if IMU y points left during level motion.

Spiral: Over 90.82 s, early angular-rate variations settle to 0.403 rad/s (mean over 25–90 s). Full-run mean $a_y = 0.0390$ m/s^2 is similarly compatible with inward acceleration under that alignment.

Line: Over 12.19 s, mean $\omega_z = 5.4 \times 10^{-4}$ rad/s indicates little net rotation. Means $a_x = 0.0017$, $a_y = -0.0010$ m/s^2 are near zero, with smaller a_y fluctuations. A small signed a_x mean does not prove constant velocity; odometry shows speed increasing.

3 LiDAR results and map acquisition

Figure 3 shows scans nearest each recording's midpoint. For 720 original beam indices, $\alpha_i = \alpha_{\min} + i\Delta\alpha$, $x_i = r_i \cos\alpha_i$, and $y_i = r_i \sin\alpha_i$. Removing infinite returns leaves 456, 458, and 488 points; finite positive ranges remain because sensor limits were not logged. Wall-like segments have different orientations in these local sensor frames.

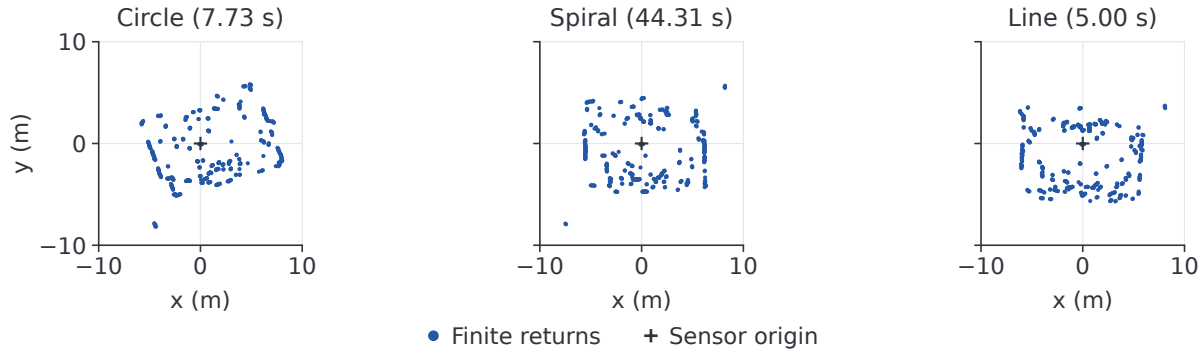


Figure 3: Midpoint LiDAR scans in their individual sensor frames; + marks the sensor origin. Times are relative to each LiDAR log's first timestamp.

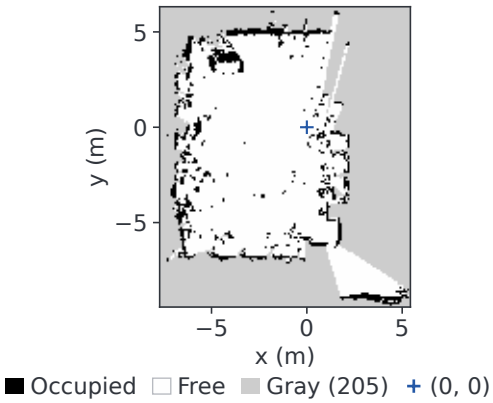


Figure 4: Occupancy-grid map of the environment, obtained from LiDAR scans using ROS 2 SLAM, visualized in RViz, and saved as a PGM file using Nav2's `map_saver_cli`. The robot was located at the bottom-right corner of the image.

Overall Interpretation:

- **Odometry:** Outputs position and heading, showing a partial circle, widening spiral loops that later overlap, and a nearly straight path with increasing speed.
- **IMU:** Outputs sensor-frame acceleration and angular rate, indicating turning during the circle and spiral, and little net turning during the line.
- **LiDAR:** Outputs distances to surrounding surfaces; the saved map shows free space and obstacles.